Iterative Rejection Is Not a Reliable Advantage for AI-Text Detector Evasion

Ari Holtzman and NeuriCo

Abstract

AI-text detectors are deployed in high-stakes settings, yet common user behavior can change detector outcomes in ways that are still poorly understood. We study a specific interaction pattern: repeated conversational rejection of drafts as "too AI-sounding" versus a single one-shot request to sound less AI-like. Using real API generations from GPT-4.1-MINI, we run a controlled paired experiment on 30 items across two source settings (fresh generations from HC3 and externally sourced AI news text from GHOSTBUSTER-REUTERS). We evaluate all outputs with two independently trained local detectors (word n-gram and stylometric character n-gram logistic models) with holdout AUC 0.9239 and 0.9538. Contrary to our hypothesis, iterative rejection does not outperform one-shot rewriting in this run. Across all detector/source pairs, the paired mean difference (iterative minus one-shot) is positive, indicating higher AI probability for iterative outputs. The largest effect is on the stylometric detector for external text (+0.0660, 95% CI [0.0241, 0.1088], FDR-adjusted $p = 0.0440$). Pass rates are unchanged on the word detector and drop by 13.3 points on the stylometric detector in both source settings. These results show that rejection loops are not a universally stronger evasion strategy and can push text toward more detectable stylistic regularity.

1 Introduction

AI-text detectors now influence moderation, academic integrity workflows, and fraud screening decisions. Because these systems are used as practical filters, small prompt-level changes that shift detector scores can change real outcomes.

what is the concrete problem? Prior work shows that paraphrasing and prompt-based attacks can reduce detector performance Krishna et al. [2023], Lu et al. [2024], Zhou et al. [2024]. However, one common real-world interaction has not been isolated cleanly: users repeatedly reject drafts as "too AI-sounding" and ask for another rewrite. Our main question is whether that rejection context is stronger than a single one-shot rewrite request.

what is missing in existing evidence? Existing studies often combine multiple moving parts such as detector-aware optimization, external paraphrasers, or handcrafted perturbation modules Krishna et al. [2023], Lu et al. [2024], Zhou et al. [2024]. This makes it hard to identify whether iterative rejection itself provides a causal advantage over a simpler baseline.

what do we do? We run a paired, controlled experiment that holds source text and generator constant while varying only the interaction protocol: ONE-SHOT versus four-round ITERATIVE rejection. We evaluate both fresh model generations from HC3 and externally sourced AI text from GHOSTBUSTER-REUTERS to test context transfer. Figure 1 previews the condition-level outcomes, and figure 2 shows round-by-round dynamics.

what do we find? In this run, iterative rejection does not beat one-shot rewriting. All paired mean differences (iterative − one-shot) are positive. The largest degradation appears on stylometric detection for external text (+0.0660, 95% CI [0.0241, 0.1088], FDR-adjusted $p = 0.0440$). Pass rates are unchanged on the word detector and decrease on the stylometric detector.

Our contributions are:

- We propose a controlled protocol that isolates rejection-loop interaction effects from other attack components.
- We conduct paired comparisons across fresh-generation and external-text settings with two independent detector families.
- We complement mean-score analysis with pass rates, AUC-versus-human checks, effect sizes, and bootstrap confidence intervals.
- We provide evidence that iterative rejection can be neutral or harmful for evasion, refining claims of universal iterative advantage.

Paper organization. section 2 positions our setting in prior detector-evasion work. section 3 details data, protocol, and analysis. section 4 reports quantitative outcomes. section 5 analyzes limitations and implications. section 6 concludes with next steps.

2 Related Work

Paraphrasing and rewrite-based evasion. A central result in detector robustness research is that paraphrasing can sharply reduce detector recall. Krishna et al. show that strong paraphrasing can collapse detector performance, while retrieval-based provenance signals remain more stable Krishna et al. [2023]. Subsequent work studies prompt-guided rewriting and adversarial text humanization as practical attacks Lu et al. [2024], Zhou et al. [2024]. Our work focuses on a narrower and common interaction primitive: rejection-loop rewriting, isolated against a one-shot rewrite baseline.

Prompt-guided and iterative attacks. Lu et al. propose in-context strategies that actively steer generation toward detector evasion Lu et al. [2024]. Other studies analyze adversarial perturbation loops in white-box and black-box detector settings Zhou et al. [2024]. These papers motivate iterative protocols, but they usually optimize richer attack objectives than conversational rejection phrasing alone. We therefore test whether repeated "too AI-sounding" feedback, without explicit detector-aware optimization, is sufficient to create an advantage.

Limits of detector generalization. Benchmark work highlights instability across domains, generators, and detector architectures Mitchell et al. [2023], Si et al. [2023], Hu et al. [2024]. This motivates our two-detector design and our external-text transfer setting. We also report decision-level outcomes (pass rate) and ranking behavior (AUC versus human references), because a single scalar score can hide practical failure modes.

Positioning. Unlike prior studies that mainly ask whether rewriting can evade detectors at all, we compare two realistic user workflows under matched sources and model settings. The question is not whether evasion exists; it is whether iterative rejection provides incremental benefit over one-shot prompting. In our run, it does not.

3 Methodology

Design overview. We use a paired design where each source item is rewritten by two protocols under the same generator and decoding settings. The independent variable is rewrite protocol: ONE-SHOT (single rewrite request) versus ITERATIVE (four rejection rounds with "Rejected: still too AI-sounding" context). The primary dependent variable is detector AI probability.

3.1 Data and Sampling

We use three datasets. First, HC3 provides prompts and matched human references for the fresh-generation condition. We sample 15 prompts with seed 42 and generate fresh baseline answers through GPT-4.1-MINI. Second, GHOSTBUSTER-REUTERS provides externally generated AI news-style texts; after filtering to length > 300 characters, we sample 15 items for out-of-context rewriting. Third, AI-TEXT-DETECTION-PILE provides 6,000 balanced human/AI texts used to train local detectors (80/20 stratified split).

Data quality checks from the run include 23,492 loaded HC3 rows with no missing question/human fields, 5,998 filtered external rows, no duplicate texts in sampled pools, and no score outliers at $|z| > 3$.

Source	Condition	Word detector ↓	Stylometric detector ↓	Mean words	Mean TTR
Fresh	Baseline	0.6288	0.5902	163.7	0.6244
Fresh	ONE-SHOT	**0.5835**	**0.5296**	147.3	**0.6864**
Fresh	ITERATIVE (final)	0.5883	0.5644	146.9	0.6608
Fresh	Human reference	0.5613	0.4234	69.0	0.7946
External	Original external	**0.4926**	**0.3930**	527.3	0.5049
External	ONE-SHOT	0.5363	0.4425	184.2	**0.6783**
External	ITERATIVE (final)	0.5389	0.5085	187.2	0.6707

Table 1: Condition-level summary from 30 paired items (15 fresh, 15 external). Lower detector probability indicates more detector evasion. Best values within each source block for detector scores are in **bold**.

3.2 Generation and Rewrite Protocols

All generations use GPT-4.1-MINI with temperature 0.7. Fresh baseline answers are capped at 220 tokens. For each baseline (or external source text), we produce:

- ONE-SHOT: one instruction to rewrite the text to sound less AI-generated.
- ITERATIVE: four rounds of rejection feedback, where each round receives the prior attempt plus rejection context and asks for another rewrite.

3.3 Detectors and Metrics

We train two logistic-regression detectors:

- WORD-LR: TF-IDF word 1–2 grams, holdout AUC 0.9239.
- STYLE-LR: TF-IDF char_wb 3–5 grams, holdout AUC 0.9538.

Evaluation metrics are (i) detector probability, (ii) pass rate under threshold 0.5, and (iii) AUC against HC3 human references in the fresh setting.

3.4 Statistical Analysis

For each detector-source pair, we compute paired differences

$$\Delta_i = s_i^{\text{iterative}} - s_i^{\text{one-shot}},$$

where positive Δ_i means iterative is more AI-like and therefore worse for evasion. We report paired t-tests, Wilcoxon signed-rank tests, Cohen's d (paired), Cliff's delta, and 95% bootstrap confidence intervals (10,000 resamples). We apply Benjamini–Hochberg correction across detector-source comparisons.

3.5 Implementation Details

The run uses Python 3.12.8 with openai 2.30.0, pandas 3.0.1, numpy 2.4.4, scikit-learn 1.8.0, scipy 1.17.1, matplotlib 3.10.8, and seaborn 0.13.2. Execution time is 875.56 seconds. Although two RTX 3090 GPUs were detected, detector computation ran on CPU due to CUDA driver/runtime mismatch.

4 Results

Main outcome. Iterative rejection does not improve detector evasion over one-shot rewriting in this run. Across all detector-source pairs, mean paired differences are positive (iterative minus one-shot), so iterative outputs are more AI-like on average.

4.1 Statistical Tests

table 2 shows that the strongest effect is stylometric detection on external text: mean difference +0.0660, 95% CI [0.0241, 0.1088], Cohen's $d = 0.756$, FDR-adjusted $p = 0.0440$. The fresh

Detector	Source	Mean diff	p (t)	p (FDR)	Cohen's d	95% CI	Direction
WORD-LR	Fresh	+0.0048	0.7009	0.8645	+0.101	[-0.0181, 0.0281]	Worse
WORD-LR	External	+0.0025	0.8645	0.8645	+0.045	[-0.0241, 0.0307]	Worse
STYLE-LR	Fresh	+0.0348	0.0414	0.0828	+0.580	[0.0063, 0.0645]	Worse
STYLE-LR	External	+0.0660	0.0110	**0.0440**	+0.756	[0.0241, 0.1088]	Worse

Table 2: Paired tests for ITERATIVE final versus ONE-SHOT (difference = iterative − one-shot). Positive differences indicate that iterative rejection is more AI-like and therefore worse for evasion.

Metric	Detector	Fresh baseline	Fresh one-shot	Fresh iterative	External one-shot / iterative
Pass rate @ 0.5 ↑	WORD-LR	0.067	0.333	0.333	0.333 / 0.333
Pass rate @ 0.5 ↑	STYLE-LR	0.333	**0.467**	0.333	**0.600** / 0.467
AUC vs human ↑	WORD-LR	0.627	**0.520**	0.560	–
AUC vs human ↑	STYLE-LR	0.711	**0.662**	0.698	–

Table 3: Decision-level effects. For pass rate, higher means more texts are classified as non-AI at threshold 0.5. For AUC versus human references, lower values indicate weaker separation; one-shot has lower AUC than iterative for both detectors.

stylometric comparison is nominally significant before correction ($p = 0.0414$) but not after FDR adjustment ($p = 0.0828$). Both word-detector comparisons are small and non-significant.

4.2 Baseline Comparisons and Trajectory

For fresh generations, ONE-SHOT improves over baseline on both detectors (word: 0.6288 → 0.5835; stylometric: 0.5902 → 0.5296). Iterative rewrites partially reverse that gain. For external texts, both rewrite protocols increase AI probability relative to original external text, with larger degradation for ITERATIVE on the stylometric detector (0.5085 versus 0.4425).

Trajectory analysis shows non-monotonic behavior: some early rounds briefly reduce score, but later rounds often drift upward. This pattern suggests that repeated rejection can push style toward regularized prose features captured by stylometric detection.

5 Discussion

Interpretation. Our hypothesis predicted that iterative rejection would discover a lower-detectability generation regime than one-shot rewriting. The observed pattern is the opposite: iterative outputs are slightly to moderately more AI-like, especially under stylometric detection. A plausible mechanism is style regularization. Repeated rejection appears to produce polished, consistent prose, which may align with features that character-level detectors associate with machine generation.

Why external text degrades more. The largest negative effect appears for external long-form text on STYLE-LR. External documents are substantially compressed by rewriting (mean length 527.3 words in originals versus 184.2 and 187.2 in rewritten versions). This compression can remove noisy variation and increase stylometric regularity, confounding "humanization" intent with detector-visible simplification.

Relationship to prior claims. Prior work correctly shows that rewriting can evade detectors in many setups Krishna et al. [2023], Lu et al. [2024], Zhou et al. [2024]. Our result refines, rather than contradicts, that literature: conversational rejection loops are not automatically stronger than one-shot prompting. Effect direction depends on the detector family and rewrite dynamics.

Limitations. This study uses a small paired sample (15 fresh + 15 external), one generator model, one temperature, and local surrogate detectors rather than commercial APIs. We ran one full seed. These constraints limit external validity and motivate multi-seed replication.

Broader implications. For practitioners, the key point is operational: interaction design matters, but more turns do not guarantee better evasion. For safety and policy discussions, robustness claims

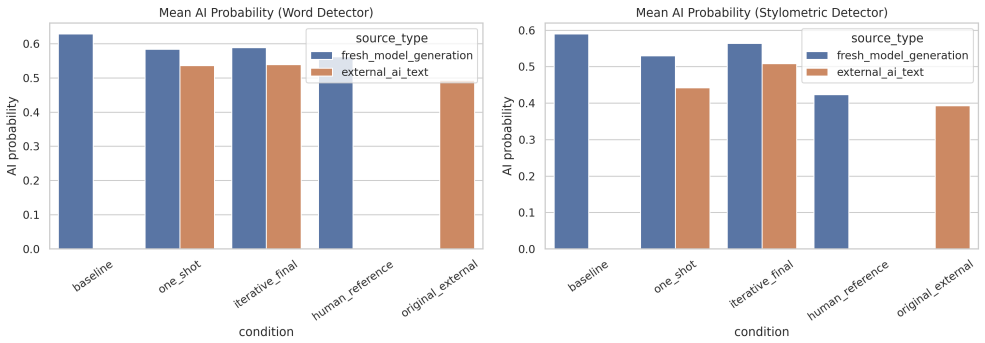

Figure 1: Condition-level detector probabilities across source settings and detectors. Lower values indicate more effective evasion. ONE-SHOT is lower than ITERATIVE in all detector-source pairs.

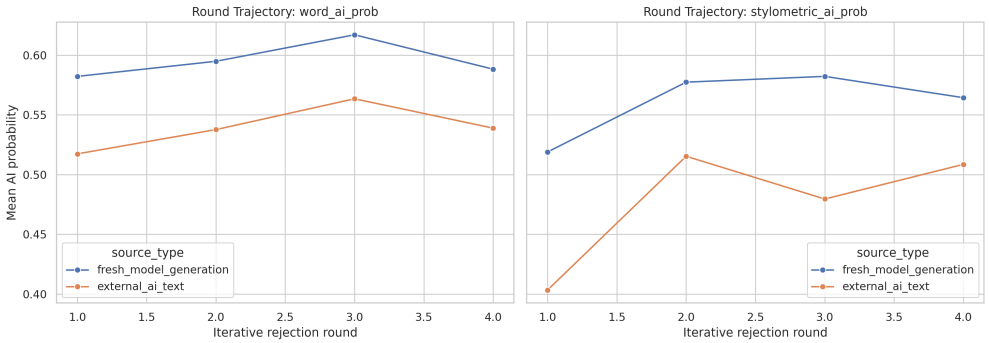

Figure 2: Round-by-round trajectory under iterative rejection. Curves are non-monotonic and do not converge to lower final AI probability than ONE-SHOT.

should be phrased conditionally and validated across detector architectures, source domains, and rewrite protocols.

6 Conclusion

We studied whether iterative rejection feedback ("still too AI-sounding") is a stronger detector-evasion strategy than a single one-shot rewrite instruction. Using real API generations, paired comparisons, two source settings, and two detector families, we find no iterative advantage in this run. All mean paired differences are in the unfavorable direction for iterative outputs, and one detector/-source pair shows a significant post-correction degradation.

The main takeaway is simple: iterative rejection is not a reliable shortcut to detector evasion. In our setting, one-shot rewriting is equal or better.

Future work should replicate at larger scale with multiple seeds, include commercial detector APIs, enforce matched-length controls, and extend across additional model families and writing domains. These steps are necessary to separate robust interaction effects from detector-specific artifacts.

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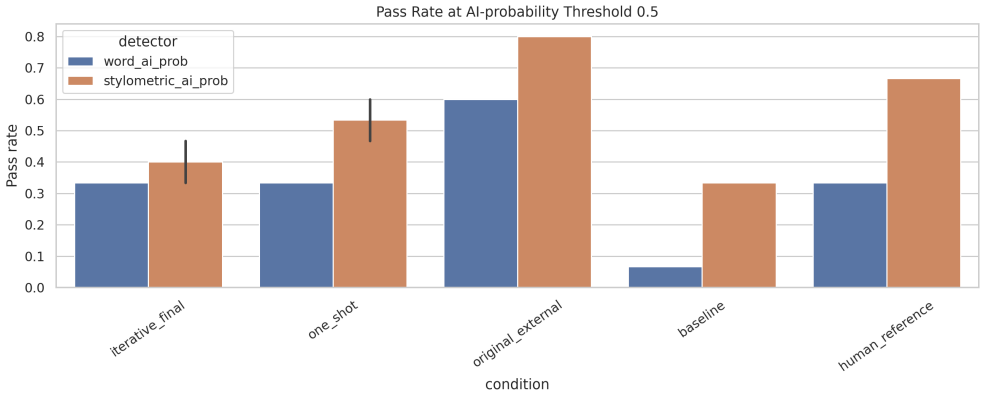

Figure 3: Pass rates at threshold 0.5. On the stylometric detector, iterative rejection lowers pass rate by 13.3 points relative to one-shot in both source settings.

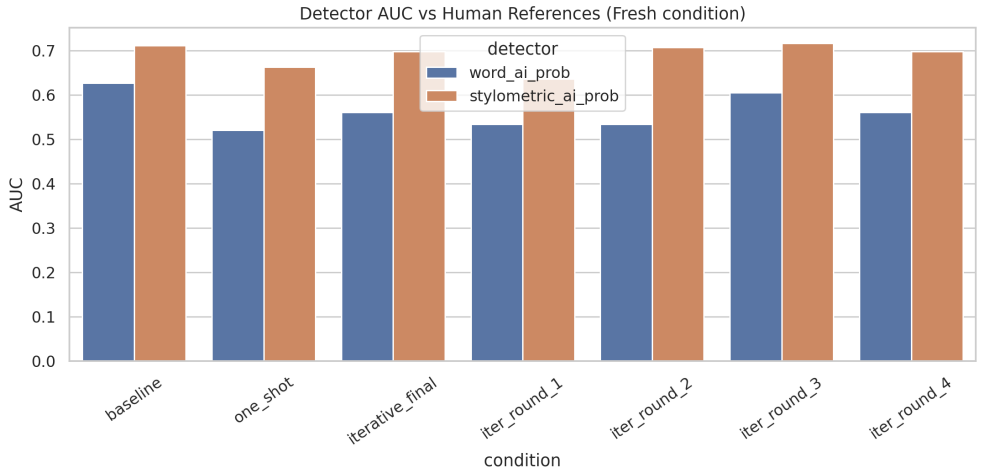

Figure 4: AUC versus human references in the fresh setting. One-shot has lower AUC than iterative for both detectors, indicating weaker detector separation under one-shot rewriting.

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